

1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).

- Input Format:
- A single floating-point number representing the mass of the object in kilograms.
 - A single floating-point number representing the velocity of the object in meters per second.

- Output Format:
- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places)

Sample Test Cases

```
1 mass = float(input())
2 velocity = float(input())
3 momentum = mass * velocity
4 print(f"momentum: {momentum:.2f}kgm/s")
```

Average time	Maximum time
0.003 s 3.25 ms	0.004 s 4.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1	Expected output	Actual output
5.8	5.8	5.8
10.8	10.8	10.8
50.00kgm/s	50.00kgm/s	50.00kgm/s

Test case 2

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1.1.2. Conditional Calculation Based on the Number of Digits

07:29

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

$1 \leq n \leq 999$

Input Format:

The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 import math
2
3 n = int(input())
4
5 if n>=1 and n<=9:
6     print(n*n)
7 elif n>=10 and n<=99:
8     print(f"{math.sqrt(n):.2f}")
9 elif n>=100 and n<=999:
10    print(f"{math.cbrt(n):.2f}")
11 else:
12    print("Invalid")
```

Average time

0.002 s

2.14 ms

Maximum time

0.003 s

3.00 ms

4 out of 4 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test Case 1

Expected output

81

Actual output

81

Test case 2

Test case 3

Terminal

Test cases

< Prev

Reset

Submit

Next >

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1.1.3. Days between Two Dates

35/40

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`.
- The second line contains the second date in the format `YYYY-MM-DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

```
1 from datetime import date
2
3 #write your code here...
4 from datetime import date
5
6 date1_str = input()
7 date2_str = input()
8
9 year1, month1, day1 = map(int, date1_str.split('-'))
10 year2, month2, day2 = map(int, date2_str.split('-'))
11
12 d1 = date(year1, month1, day1)
13 d2 = date(year2, month2, day2)
14
15 delta = d2 - d1
16 print(delta.days)
```

Average time

0.002 s

2.25 ms

Maximum time

0.003 s

3.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

3.10s

Debug

Expected output

2014-07-02

2014-11-02

123

Actual output

2014-07-02

2014-11-02

123

Test case 2

4.1ms

Terminal

Test cases

< Prev

Reset

Submit

Next >

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 #write your code here...
2 num=int(input())
3 reversed_num=int(str(num)[::-1])
4 print(reversed_num)
```

Average time0.002 s2.20 msMaximum time0.003 s3.00 ms

2 out of 2 shown test case(s) passed3 out of 3 hidden test case(s) passed

Test case 12.00sDebug

Expected output53677635Actual output53677635

Test case 22.00s

TerminalTest cases

PrevResetSubmitNext

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1.1.5. Multiplication Table

01:45

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

The character "x" is used in the output to represent multiplication.
Refer to the visible test-cases for more clarity.

Sample Test Cases

multipl...

```
1 #write your code here...
2 num=int(input())
3 for i in range(1,11):
4     print(f"{num} x {i} = {num*i}")
```

Average time0.001 s1.25 ms

Maximum time0.002 s2.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 12.00sDebug

Expected output

0 x 1 = 0
0 x 2 = 16
0 x 3 = 24
0 x 4 = 32
0 x 5 = 40

Actual output

0
0 x 1 = 0
0 x 2 = 16
0 x 3 = 24
0 x 4 = 32
0 x 5 = 40

TerminalTest cases

PrevResetSubmitNext

1.2.1. Pass or Fail

0122

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

Submit

1

write your code here

2

n = int(input())

3

marks = list(map(int, input().split()))

4

Check fail condition

5

failed = False

6

for m in marks:

7

if m < 40:

8

failed = True

9

break

10

if failed:

11

print("Fail")

12

else:

13

avg = sum(marks) / n

14

print(f"Aggregate Percentage: {avg:.2f}")

15

16

if avg > 75:

17

print("Grade: Distinction")

18

elif avg >= 60:

19

print("Grade: First Division")

20

elif avg >= 50:

21

print("Grade: Second Division")

22

else:

23

print("Grade: Third Division")

Average time

0.003 s

2.70 ms

Maximum time

0.004 s

4.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1

4 ms

Debug

Expected output

4

55 65 70 89

Aggregate Percentage: 71.75

Grade: First Division

Actual output

4

55 65 70 89

Aggregate Percentage: 71.75

Grade: First Division

Test case 2

Terminal

Test cases

< Prev

Reset

Submit

Next >

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     if n == 1:
3         return 0
4     if n == 2:
5         return 1
6     else:
7         return fibonacci(n-1)+fibonacci(n-2)
8
9 n = int(input())
10 for i in range(1, n+1):
11     print(fibonacci(i), end=" ")
12
```

Average time0.006 s6.25 msMaximum time0.018 s18.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 14 msDebug

Expected output50 1 1 2 3Actual output50 1 1 2 3

Test case 24 ms

TerminalTest cases

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format

The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 #Type Content here...
2 n = int(input())
3
4 for i in range(1, n + 1):
5     print("*" * i)
```

Average time0.001 s1.33 ms

Maximum time0.002 s2.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1

Expected output

Actual output

5

*
**

5
*
**

Terminal

Test cases

Prev

Reset

Submit

Next

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1.2.4. Pattern - 205/22

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 # Type: Content here...
2 # Input number of rows
3 n = int(input())
4
5 # Loop through rows
6 for i in range(1, n + 1):
7     for j in range(1, i + 1):
8         print(j, end=" ")
9     print()
```

Average time0.002 s2.00 ms

Maximum time0.003 s3.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 13/100Debug

Expected output

5

1 1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Actual output

5

1 1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

TerminalTest cases

Network: 7Internet access

SubmitNext